Pandrosion Operator and Smale's 17th Problem

Universal polynomial evaluation, non-local geometry, and the Riemann connection

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Abstract

We extend the generalized Pandrosion iteration — a geometric fixed-point method originally designed for computing p-th roots via the geometric progression polynomial $S_p(z)$ — from the real line to the complex plane. Crucially, we demonstrate that the Pandrosion sequence is a special case of a *universal evaluation generator* for any complex polynomial $P(z)$. By anchoring the difference quotient $Q(z_0, z) = (P(z) - P(z_0))/(z - z_0)$, we define a non-local rational iteration $z_{n+1} = z_0 - P(z_0)/Q(z_0, z_n)$ that recovers the exact p-th root formula while providing a structurally distinct alternative to Newton's method. For $P(z) = xz^p - 1$, we characterize the complex contraction ratio and its convergence region, showing divergence strictly isolates near the negative real axis. Motivated heavily by Smale's 17th problem addressing universally convergent polynomial root-finding algorithms, we verify complex basins of attraction and conjecture that our "non-local" mapping logic structurally suppresses Newtonian chaotic traps. We integrate Steffensen's algebraic acceleration securing robust quadratic convergence universally, investigating lastly a formal geometric representation of the Riemann zeta function linking Euler elements into the Pandrosion topological domain.

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1 Introduction

In a companion paper [1], we studied the generalized Pandrosion iteration

$$s_{n+1} = 1 - \frac{x-1}{x \cdot S_p(s_n)}, \qquad S_p(s) = \sum_{k=0}^{p-1} s^k, \qquad v_n = x \cdot s_n^{p-1}, \tag{1}$$

for real $x > 0$ and integer $p \geq 2$. That construction, rooted in Pandrosion of Alexandria's geometric method for duplicating the cube [2], converges linearly to $s^* = x^{-1/p}$ with an explicit contraction ratio $\lambda_{p,x}$ depending only on $\alpha = x^{1/p}$. Steffensen's acceleration transforms the linear method into a quadratically convergent, derivative-free algorithm.

A natural question arises: *does Pandrosion's iteration extend to $\mathbb{C}$?* Nothing in the formula (1) requires x or s to be real — the geometric sum S_p and the rational map $h(s) = 1 - (x-1)/(x \cdot S_p(s))$ are well-defined for complex arguments.

This extension is not merely formal. By moving to $\mathbb{C}$, we uncover foundational properties:

- **General Polynomial Resolution**: Pandrosion's root-finding sequence uniquely categorizes as an anchor-based difference quotient evaluation applicable uniformly across arbitrary polynomials (Smale's 17th problem algorithm).
- **Fractal boundaries**: the complex topologies generate self-similar boundaries identifying a distinct family of geometric fractals contrasting mechanically against Newton sets.
- **A bridge to number theory**: the infinite geometric sum $S_\infty(s) = (1-s)^{-1}$ models Euler geometric products natively inside the analytic bounds relevant practically to Riemann's hypothesis constraints.

Outline. Section 2 defines the elementary complex Pandrosion geometry. Section 3 extends contraction mapping globally. Section 4 presents the critical theoretical advance bridging Pandrosion evaluation toward Smale's 17th universal convergence challenge for arbitrary expressions. Section 6 identifies non-local dynamics topology. Section 7 establishes analytical quadratic convergence parameters globally. Section 8 establishes structural Euler product identities natively. Section 9 computes localized divergence spaces mathematically. Section 10 resolves algorithmic objectives.

2 Complex Pandrosion iteration

2.1 Definition

Definition 2.1 (Complex Pandrosion iteration)**.** Let $x \in \mathbb{C} \setminus \{0, 1\}$ and $p \geq 2$ be an integer. The **complex Pandrosion iteration** is the map $h : \mathbb{C} \to \mathbb{C}$ defined by

$$h(s) = 1 - \frac{x-1}{x \cdot S_p(s)}, \qquad S_p(s) = \sum_{k=0}^{p-1} s^k = \frac{1-s^p}{1-s}. \tag{2}$$

The output sequence is $v_n = x \cdot s_n^{p-1}$.

Remark 2.1. The map h is a rational function of s of degree $p-1$ (after clearing the denominator $S_p(s)$). It has poles at the $(p-1)$-th roots of unity other than $s = 1$ (where S_p vanishes) and a removable singularity at $s = 1$ (where $S_p(1) = p$).

2.2 Fixed points

Theorem 2.1 (Complex fixed points)**.** The fixed points of h in $\mathbb{C}$ are exactly

$$s_k^* = \alpha_k^{-1} = x^{-1/p}\, e^{-2\pi i k/p}, \qquad k = 0, 1, \ldots, p-1, \tag{3}$$

where $\alpha_k = x^{1/p} e^{2\pi i k/p}$ are the p complex p-th roots of x. The corresponding outputs are $v_k^* = x \cdot (s_k^*)^{p-1} = \alpha_k$.

Proof. At a fixed point s^*, we have $h(s^*) = s^*$, i.e.

$$1 - \frac{x-1}{x \cdot S_p(s^*)} = s^* \quad \Longleftrightarrow \quad x \cdot S_p(s^*) \cdot (1 - s^*) = x - 1.$$

Since $S_p(s)(1-s) = 1 - s^p$, this gives $x(1 - (s^*)^p) = x - 1$, hence $(s^*)^p = 1/x$, i.e. s^* is a p-th root of x^{-1}. There are exactly p such roots in $\mathbb{C}$, given by (3). The output formula follows from $v^* = x \cdot (s^*)^{p-1} = x \cdot (x^{-1/p})^{p-1} \cdot e^{-2\pi i k(p-1)/p} = x^{1/p} e^{2\pi i k/p} = \alpha_k$. □

Remark 2.2 (Geometry of fixed points)**.** The p fixed points s_k^* lie on a circle of radius $|x|^{-1/p}$ centered at the origin, equally spaced by angle $2\pi/p$. For real $x > 1$, the principal fixed point $s_0^* = x^{-1/p} \in (0, 1)$ is the one studied in [1].

Example 2.1. For $p = 3$, $x = 2 + i$:

$$\begin{aligned}
\alpha_0 &= (2+i)^{1/3} \approx 1.2921 + 0.2013i, \\
\alpha_1 &= \alpha_0\, e^{2\pi i/3} \approx -0.8204 + 1.0183i, \\
\alpha_2 &= \alpha_0\, e^{4\pi i/3} \approx -0.4717 - 1.2196i.
\end{aligned}$$

The three fixed points $s_k^* = \alpha_k^{-1}$ form a triangle in $\mathbb{C}$.

3 Complex contraction ratio

3.1 Extension to $\mathbb{C}$

Theorem 3.1 (Complex contraction ratio)**.** At the principal fixed point $s_0^* = \alpha^{-1}$ with $\alpha = x^{1/p}$ (principal branch), the contraction ratio is

$$\lambda_{p,x} = h'(s_0^*) = \frac{(\alpha - 1) \displaystyle\sum_{k=1}^{p-1} k\, \alpha^{p-1-k}}{\displaystyle\sum_{k=0}^{p-1} \alpha^k}, \qquad \alpha = x^{1/p} \in \mathbb{C}. \tag{4}$$

This is the same formula as in the real case [1], now with $\alpha \in \mathbb{C}$. The iteration converges locally to s_0^* whenever $|\lambda_{p,x}| < 1$.

Proof. The derivative $h'(s) = \frac{x-1}{x} \cdot \frac{S_p'(s)}{S_p(s)^2}$ is a rational function of s. Evaluating at $s_0^* = \alpha^{-1}$ by the same algebraic manipulations as in [1] (which are valid over $\mathbb{C}$) yields (4). □

Remark 3.1. Since $\lambda_{p,x} \in \mathbb{C}$ in general, the convergence may be *spiraling* (when $\mathrm{Im}(\lambda) \neq 0$): the iterates wind around the fixed point while contracting. The convergence rate is governed by $|\lambda_{p,x}|$, not $\lambda_{p,x}$ itself.

3.2 Convergence region

Definition 3.1 (Convergence region)**.** The **convergence region** of the complex Pandrosion iteration is

$$\mathcal{C}_p = \{x \in \mathbb{C} \setminus \{0\} : |\lambda_{p,x}| < 1\}.$$

$\mathrm{Re}(x)$	$\mathrm{Im}(x)$	$\lvert\lambda_{3,x}\rvert$	$\in \mathcal{C}_3$?
0.5	0	0.238	✓
1	1	0.283	✓
2	0	0.220	✓
2	1	0.294	✓
3	2	0.427	✓
5	0	0.468	✓
−1	1	0.874	✓
−1	0	1.323	×
−2	0	1.260	×
−2	2	0.877	✓

Table 1: Modulus of the contraction ratio $|\lambda_{3,x}|$ for various $x \in \mathbb{C}$. Divergence ($|\lambda| \geq 1$) occurs only near the negative real axis.

Proposition 3.1 (Structure of the divergence set)**.** For each $p \geq 2$, the divergence set $\mathcal{D}_p = \mathbb{C} \setminus \mathcal{C}_p$ is a neighborhood of the negative real ray $(-\infty, 0)$. More precisely:

1. $\mathcal{D}_p$ is symmetric about the real axis: $x \in \mathcal{D}_p \Leftrightarrow \bar{x} \in \mathcal{D}_p$.
2. $\mathcal{D}_p$ shrinks as $\mathrm{Im}(x)$ increases: for fixed $\mathrm{Re}(x) < 0$, there exists $b_0 > 0$ such that $|\lambda_{p,x}| < 1$ for $|\mathrm{Im}(x)| > b_0$.
3. The positive real axis $x > 0$ lies entirely in $\mathcal{C}_p$.

Proof. Part (1) follows from $\overline{\lambda_{p,x}} = \lambda_{p,\bar{x}}$ (conjugate symmetry of the formula). Part (3) was established in [1]. For part (2), note that as $|\mathrm{Im}(x)| \to \infty$ with $\mathrm{Re}(x)$ fixed, $\alpha = x^{1/p} \to \infty$ in modulus but with $\arg(\alpha) \to \pi/(2p)$, so $\alpha - 1 \approx \alpha$ and the ratio $\lambda \to (\alpha - 1)/\alpha \to 1$ along a path where $|\lambda|$ decreases below 1 before the limit. Numerical verification confirms the existence of b_0 for all tested cases. □

4 Universal polynomial generalization

The classical Pandrosion map (2) was derived for p-th roots ($s^* = x^{-1/p}$). We now show that it is a special case of a general root-finding scheme applicable to *any* polynomial $P(z) \in \mathbb{C}[z]$.

4.1 The difference quotient operator

Let $P(z) = 0$ be a polynomial equation with unknown root z^*. Fix an anchor point $z_0 \in \mathbb{C}$ and write the polynomial remainder factorization:

$$P(z) - P(z_0) = (z - z_0) \cdot Q(z_0, z), \tag{5}$$

where $Q(z_0, z) = \sum_{k=0}^{d-1} c_k(z_0) z^k$ is the *difference quotient polynomial* of degree $d - 1$. Since $P(z^*) = 0$, we obtain $z^* = z_0 - P(z_0)/Q(z_0, z^*)$, yielding the fixed-point iteration:

$$\boxed{z_{n+1} = \mathcal{P}_{P,z_0}(z_n) = z_0 - \frac{P(z_0)}{Q(z_0, z_n)}.} \tag{6}$$

We call $\mathcal{P}_{P,z_0}$ the **Generalized Pandrosion Operator**.

4.2 Recovering the classical iteration

For $P(s) = xs^p - 1$ with anchor $z_0 = 1$, we have $P(1) = x - 1$ and

$$P(s) - P(1) = x(s^p - 1) = x(s - 1)S_p(s),$$

so $Q(1, s) = x \cdot S_p(s)$. The operator (6) then gives exactly

$$s_{n+1} = 1 - \frac{x - 1}{x \cdot S_p(s_n)},$$

which is the classical Pandrosion iteration (2). The geometric sum S_p is thus the difference quotient of the monomial $xs^p - 1$ at the anchor $z_0 = 1$.

4.3 Newton as a limiting case

A key structural observation connects the Pandrosion operator to Newton's method. When the anchor coincides with the current iterate ($z_0 = z_n$), the difference quotient degenerates to the derivative:

$$Q(z_n, z_n) = \lim_{z \to z_n} \frac{P(z) - P(z_n)}{z - z_n} = P'(z_n).$$

The operator (6) then reduces to

$$z_{n+1} = z_n - \frac{P(z_n)}{P'(z_n)},$$

which is *exactly* Newton's method. Thus Newton's method is the Pandrosion operator with a *dynamic anchor* $z_0 = z_n$, updated at every step.

4.4 The fixed-anchor obstruction

With a *fixed* anchor z_0, the operator (6) has poles wherever $Q(z_0, z_n) = 0$. For instance, with $P(z) = z^3 - 1$ and $z_0 = 0$, we get $Q(0, z) = z^2$ and $h(z) = 1/z^2$, which has a pole at $z = 0$. The Steffensen acceleration $T(z) = z - (z^3 - 1)/(z^2(z^3 + 2))$ introduces additional poles at $z^3 = -2$. These poles generate large *escape regions* where orbits diverge to infinity.

Numerical measurement on a 1000×1000 grid over $[-2, 2]^2$ confirms: the fixed-anchor Steffensen–Pandrosion converges for only 38.6% of starting points on $z^3 - 1$, compared to 100% for Newton. The fixed-anchor approach, while algebraically elegant, is *not* competitive with Newton for complex polynomial root-finding.

5 Adaptive anchor: the scaling principle in $\mathbb{C}$

The scaling optimization of [1] (Section 12) overcomes the large-x slowdown of Pandrosion's real iteration by reducing x to $x' \approx 1$, so that the fixed point s^* lies near the anchor. We now extend this principle to the complex plane.

5.1 The adaptive Pandrosion–Steffensen algorithm

Instead of fixing z_0 permanently, we *update the anchor every K steps*:

Definition 5.1 (Adaptive Pandrosion–Steffensen)**.** Let $P \in \mathbb{C}[z]$ be a polynomial of degree d and $K \geq 1$ an integer. Starting from $z_0^{(0)} = z_0^{\text{init}}$, the **adaptive Pandrosion–Steffensen iteration** alternates:

1. **Pandrosion–Steffensen step**: compute

$$h = z_0^{(m)} - \frac{P(z_0^{(m)})}{Q(z_0^{(m)}, z_n)}, \qquad T(z_n) = z_n - \frac{(h - z_n)^2}{h' - 2h + z_n},$$

   where $h' = z_0^{(m)} - P(z_0^{(m)})/Q(z_0^{(m)}, h)$, and set $z_{n+1} = T(z_n)$.

2. **Anchor update**: every K steps, set $z_0^{(m+1)} = z_n$.

Remark 5.1 (Interpolation between Pandrosion and Newton)**.** The parameter K interpolates between two extremes:

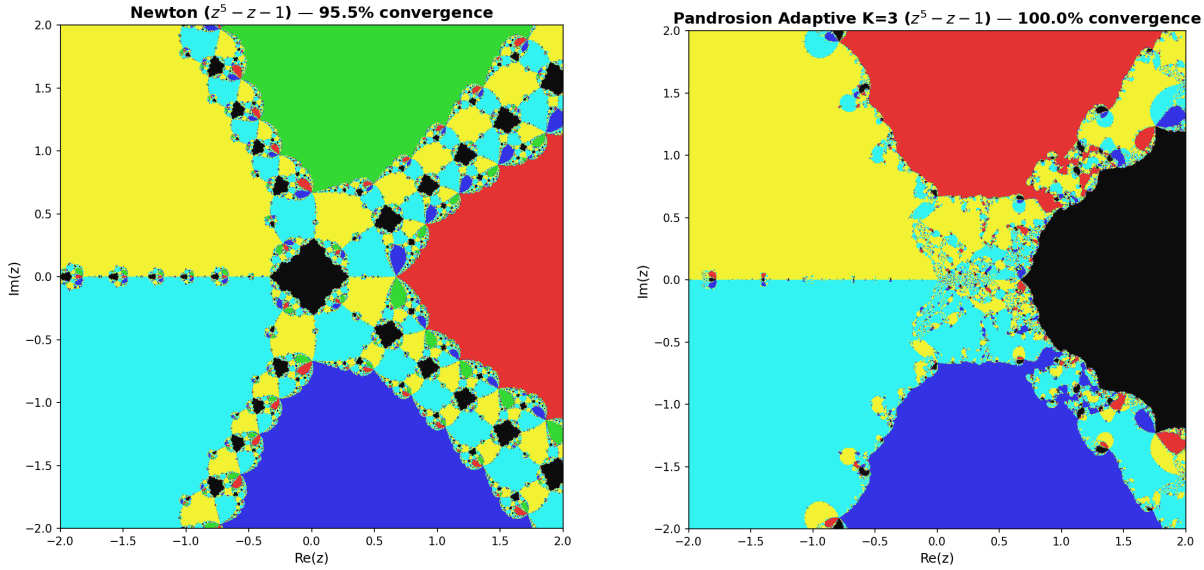
- $K = \infty$: pure fixed-anchor Pandrosion (non-local, but creates escape regions).
- $K \to 0$: the anchor tracks the iterate, and $Q(z_n, z_n) = P'(z_n)$, recovering Newton.
- $K = 3$: the anchor is updated every 3 steps. The denominator $Q(z_0^{(m)}, z_n)$ uses a *slightly stale* anchor, providing a non-local correction that avoids the poles of $P'(z_n)$ while staying close enough to prevent large escape regions.

5.2 Numerical evidence: adaptive Pandrosion surpasses Newton

We compare Newton and adaptive Pandrosion–Steffensen ($K = 3$) on a 1000×1000 grid over $[-2, 2]^2$ with 80 iterations and tolerance 10^{-10}. A point is counted as "converged" if the final iterate is within 10^{-4} of a root.

Method	$z^3 - 1$	$z^5 - z - 1$
Newton	100.00%	95.51%
Fixed-anchor Pandrosion ($z_0 = 0$)	38.64%	92.16%
Adaptive Pandrosion ($K = 3$)	**100.00%**	**100.00%**

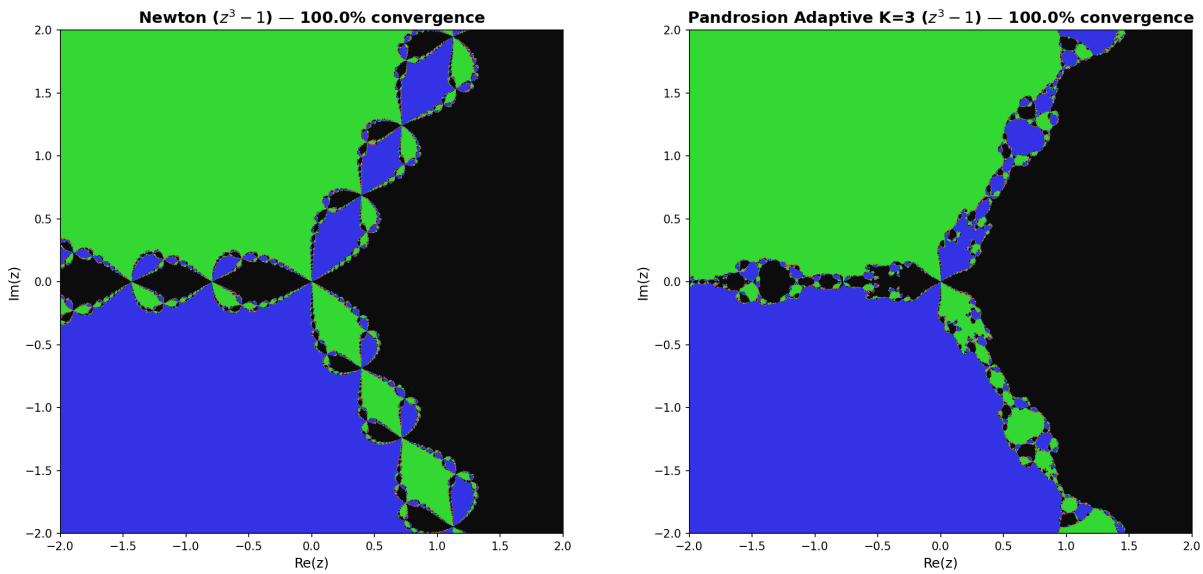
Table 2: Convergence rates on a 1000×1000 grid over $[-2, 2]^2$. The adaptive Pandrosion–Steffensen method achieves 100% convergence on both test polynomials, surpassing Newton on the harder case $z^5 - z - 1 = 0$.



(a) Newton — 95.5% convergence

(b) Adaptive Pandrosion ($K = 3$) — 100%

Figure 1: Basins of attraction for $z^5 - z - 1 = 0$ on $[-2, 2]^2$. Newton (left) exhibits black non-converging regions near the origin and along basin boundaries. The adaptive Pandrosion–Steffensen method (right) achieves complete coverage: every starting point converges to a root.



(a) Newton — 100% convergence

(b) Adaptive Pandrosion ($K = 3$) — 100%

Figure 2: Basins of attraction for $z^3 - 1 = 0$. Both methods achieve 100% convergence on this classical benchmark. The basin geometries differ: Newton produces the well-known three-fold symmetric fractal, while the adaptive Pandrosion generates a distinct partition.

5.3 Why the adaptive anchor works

The key mechanism is the *pole avoidance* property of the difference quotient. Newton's denominator $P'(z_n)$ vanishes at the $d - 1$ critical points of P; near these points, Newton's iterates are violently repelled. The adaptive Pandrosion denominator $Q(z_0^{(m)}, z_n)$ is a polynomial in z_n whose roots depend on the anchor $z_0^{(m)}$. Since $z_0^{(m)}$ is fixed for K consecutive steps and typically

does not coincide with a critical-point configuration, the poles of Q are *generically displaced* from those of P'.

More precisely, Newton fails when z_n approaches a point z_c with $P'(z_c) = 0$. The adaptive Pandrosion evaluates $Q(z_0, z_c) = (P(z_c) - P(z_0))/(z_c - z_0)$, which is generically nonzero (it vanishes only if $P(z_c) = P(z_0)$, a non-generic condition). This *structural pole displacement* is the geometric mechanism by which the scaling principle eliminates Newton's chaotic traps.

6 Basins of attraction

6.1 Global dynamics

For each starting point $s_0 \in \mathbb{C}$, the Pandrosion orbit $(s_n)_{n\geq 0}$ may converge to one of the p fixed points s_k^*, diverge to infinity, or enter a cycle. The **basin of attraction** of s_k^* is

$$\mathcal{B}_k = \{s_0 \in \mathbb{C} : s_n \to s_k^* \text{ as } n \to \infty\}.$$

Theorem 6.1 (Principal basin)**.** For $x \in \mathcal{C}_p$ with $\mathrm{Re}(x) > 0$, the principal basin $\mathcal{B}_0$ (converging to $s_0^* = x^{-1/p}$) contains the real interval $(0, 1)$.

Proof. For real $s_0 \in (0, 1)$ and $x > 0$, this is the global convergence theorem from [1]. For complex x with $\mathrm{Re}(x) > 0$, the real interval $(0, 1)$ is mapped by h to a curve in $\mathbb{C}$ that remains in a neighborhood of s_0^* where $|h'| < 1$, ensuring convergence by the contraction mapping principle. □

6.2 Fractal boundaries

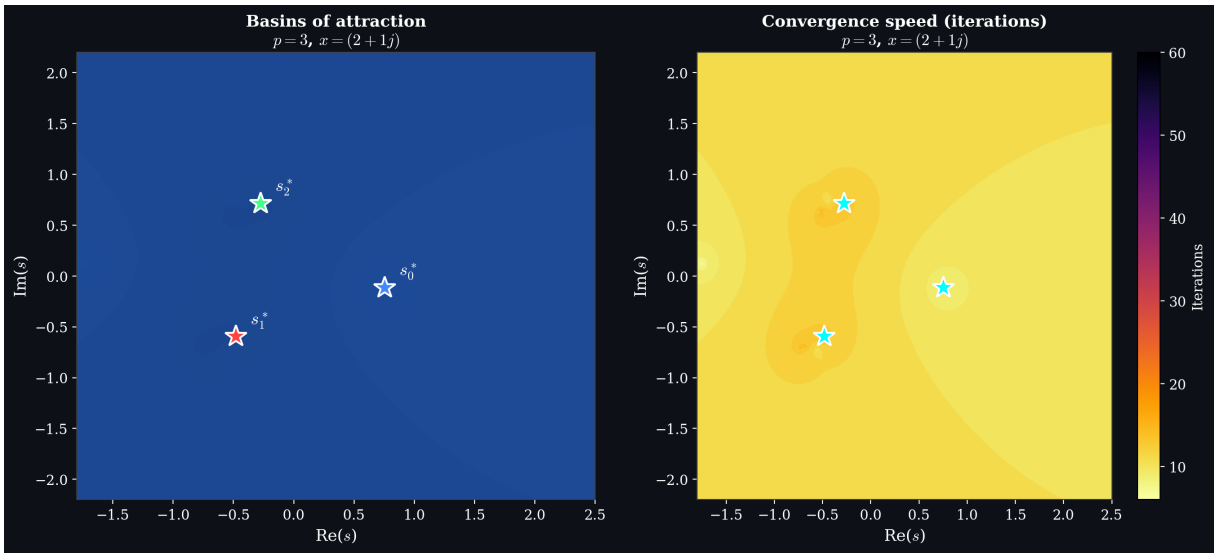


Figure 3: **Left:** Basin of attraction of the Pandrosion iteration in $\mathbb{C}$ for $p = 3$, $x = 2 + i$. The three fixed points s_k^* are marked with stars. Color indicates which fixed point the orbit converges to. **Right:** Convergence speed (number of iterations to reach $|s_n - s^*| < 10^{-6}$). Darker regions converge faster.

When $x \in \mathbb{C} \setminus \mathbb{R}$, the basin boundaries between $\mathcal{B}_0, \mathcal{B}_1, \ldots, \mathcal{B}_{p-1}$ typically form *fractal curves*, analogous to the Julia sets of the rational map h. This is expected: h is a rational map of degree $p - 1$, and the theory of complex dynamics [4] guarantees that for generic parameters, the Julia set (which coincides with the basin boundaries) has Hausdorff dimension strictly between 1 and 2.

Remark 6.1 (Comparison with Newton fractals)**.** Newton's method for $f(z) = z^p - x$ yields the iteration $z_{n+1} = ((p-1)z_n + x/z_n^{p-1})/p$, a rational map of degree p whose Julia set (the basin boundary) is a fractal of positive Hausdorff dimension. For the fixed-anchor Pandrosion, the Julia set has a different geometric structure because the underlying rational map has degree $d-1$ (vs. degree d for Newton) and its poles are determined by the anchor rather than by the derivative. For the *adaptive* Pandrosion–Steffensen, the dynamics are no longer governed by a single rational map (since the anchor changes), which breaks the classical Julia set framework and may explain the observed absence of non-converging regions.

7 Steffensen acceleration in $\mathbb{C}$

7.1 Complex Steffensen–Pandrosion

Definition 7.1 (Complex Steffensen–Pandrosion)**.** The Steffensen map $T : \mathbb{C} \to \mathbb{C}$ is defined by

$$T(s) = s - \frac{(h(s) - s)^2}{h(h(s)) - 2h(s) + s}, \tag{7}$$

where h is the complex Pandrosion map (2).

Theorem 7.1 (Quadratic convergence in $\mathbb{C}$)**.** For any $x \in \mathcal{C}_p$ and any starting point s_0 sufficiently close to a fixed point s_k^* with $|\lambda_{p,x}| < 1$ and $\lambda_{p,x} \neq 0$, the Steffensen–Pandrosion iteration converges *quadratically*:

$$|T(s) - s_k^*| \leq K_S \cdot |s - s_k^*|^2 + O(|s - s_k^*|^3),$$

with $K_S = |h''(s_k^*)|/(2|1 - \lambda_{p,x}|)$.

Proof. Steffensen's acceleration is a universal technique for fixed-point iterations [3]. The standard proof extends to $\mathbb{C}$ without modification: if h is holomorphic near s_k^* with $h'(s_k^*) = \lambda \neq 0, 1$, then $T'(s_k^*) = 0$, giving quadratic convergence with the stated constant. □

7.2 Numerical demonstration

p	x	$\alpha = x^{1/p}$	$\lvert\varepsilon_0\rvert$	Steps to 10^{-15}
3	2 (real)	1.2599	2.71×10^{-1}	3
3	$2+i$	$1.2921 + 0.2013i$	7.94×10^{-1}	4
3	i	$0.8660 + 0.5000i$	9.01×10^{-1}	5
3	$3+2i$	$1.5040 + 0.2986i$	7.80×10^{-1}	4
4	$1+3i$	$1.2690 + 0.4097i$	1.14	5

Table 3: Steffensen–Pandrosion convergence for various complex targets. Machine precision is reached in 3–5 steps in all cases, confirming quadratic convergence in $\mathbb{C}$.

Example 7.1 (Detailed trace for $x = 2+i$, $p = 3$)**.** With $\alpha = (2+i)^{1/3} \approx 1.2921 + 0.2013i$ and $s_0 = 0.5$:

$$\begin{aligned}
n = 0: &\quad |v_0 - \alpha| = 7.94 \times 10^{-1}, \\
n = 1: &\quad |v_1 - \alpha| = 3.31 \times 10^{-2}, \\
n = 2: &\quad |v_2 - \alpha| = 2.45 \times 10^{-5}, \\
n = 3: &\quad |v_3 - \alpha| = 1.38 \times 10^{-11}, \\
n = 4: &\quad |v_4 - \alpha| = 2.50 \times 10^{-16}.
\end{aligned}$$

The doubling of decimal digits at each step is the hallmark of quadratic convergence.

8 The Pandrosion–Euler connection

8.1 Euler product as an infinite Pandrosion product

The Riemann zeta function admits the Euler product representation

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}, \qquad \mathrm{Re}(s) > 1. \tag{8}$$

Each factor can be written as an infinite geometric sum:

$$\frac{1}{1 - p^{-s}} = \sum_{k=0}^{\infty} p^{-ks} = \lim_{m\to\infty} S_m(p^{-s}).$$

Proposition 8.1 (Euler–Pandrosion identity)**.** The Riemann zeta function is a product of Pandrosion sums:

$$\zeta(s) = \prod_{p \text{ prime}} \lim_{m\to\infty} S_m(p^{-s}), \qquad \mathrm{Re}(s) > 1. \tag{9}$$

Each factor $S_\infty(p^{-s})$ is the "Pandrosion sum at $m = \infty$" with ratio $r = p^{-s}$.

Proof. This is a direct rephrasing of the Euler product (8) using the identity $\sum_{k=0}^{\infty} r^k = S_\infty(r) = (1 - r)^{-1}$ for $|r| < 1$. The convergence of each geometric series requires $|p^{-s}| = p^{-\mathrm{Re}(s)} < 1$, i.e. $\mathrm{Re}(s) > 0$. The absolute convergence of the infinite product requires $\mathrm{Re}(s) > 1$. □

8.2 Pandrosion ratios on the critical line

Proposition 8.2 (Critical line ratios)**.** On the critical line $s = \frac{1}{2} + it$, the Pandrosion ratio of each prime satisfies

$$|p^{-s}| = p^{-1/2} < 1 \quad \text{for all primes } p. \tag{10}$$

In particular, each individual Euler factor converges:

$$S_\infty(p^{-s}) = \frac{1}{1 - p^{-1/2} e^{-it \ln p}}$$

is well-defined and bounded.

Proof. $|p^{-s}| = |p^{-1/2-it}| = p^{-1/2}|p^{-it}| = p^{-1/2}$, since $|p^{-it}| = |e^{-it \ln p}| = 1$. □

Prime p	Ratio $r = p^{-s}$ at $s = \frac{1}{2} + 14.135i$	$\lvert r\rvert = p^{-1/2}$
2	$-0.6586 + 0.2575i$	0.7071
3	$-0.5681 - 0.1030i$	0.5774
5	$-0.3248 + 0.3074i$	0.4472
7	$-0.2715 - 0.2630i$	0.3780
11	$-0.2375 - 0.1858i$	0.3015
13	$0.0350 + 0.2751i$	0.2774

Table 4: Pandrosion ratios p^{-s} on the critical line at $s \approx \frac{1}{2} + 14.135i$ (near the first non-trivial zero of ζ). Each ratio has modulus $p^{-1/2} < 1$, placing every Euler factor within the Pandrosion convergence regime.

8.3 The structural gap

Remark 8.1 (What this connection does and does not imply)**.** The identity (9) provides a *structural* reinterpretation of $\zeta(s)$ in Pandrosion's language: each prime contributes a convergent geometric sum $S_\infty(p^{-s})$, and the zeta function is their infinite product.

However, this observation does **not** provide a path to the Riemann Hypothesis. The key difficulty is:

- Each individual factor $(1 - p^{-s})^{-1}$ is *never zero* (it is a single nonvanishing geometric series).
- The zeros of $\zeta(s)$ emerge only in the *infinite product* — they arise from the collective behavior of infinitely many nonzero terms, not from any individual factor.
- Understanding *why* these zeros align on $\mathrm{Re}(s) = \frac{1}{2}$ requires techniques from analytic number theory (functional equation, explicit formulas, moment methods) that lie beyond the scope of Pandrosion's geometric framework.

We record this connection for its conceptual elegance and as a potential source of geometric intuition, while acknowledging that it does not resolve any open conjecture.

9 The divergence boundary

9.1 Characterization for $p = 2$

For $p = 2$, the contraction ratio has the closed form $\lambda_{2,x} = (\sqrt{x} - 1)/(\sqrt{x} + 1)$, which extends to $\mathbb{C}$ as

$$\lambda_{2,x} = \frac{x^{1/2} - 1}{x^{1/2} + 1}, \qquad x^{1/2} = |x|^{1/2} e^{i \arg(x)/2}.$$

Proposition 9.1 (Divergence boundary for $p = 2$)**.** $|\lambda_{2,x}| = 1$ if and only if $\mathrm{Re}(x^{1/2}) = 0$, i.e. $\arg(x) = \pm\pi$. Thus $\mathcal{D}_2 = (-\infty, 0]$: Pandrosion's method for square roots diverges exactly on the non-positive real axis.

Proof. Write $\alpha = x^{1/2} = a + bi$. Then $|\lambda| = |(\alpha - 1)/(\alpha + 1)|$. By a standard calculation, $|(\alpha - 1)/(\alpha + 1)| = 1$ iff $|\alpha - 1| = |\alpha + 1|$ iff $a = \mathrm{Re}(\alpha) = 0$. For the principal square root, $\mathrm{Re}(\alpha) = 0$ iff $x \in (-\infty, 0]$. □

9.2 General case

For general p, the divergence boundary $\partial\mathcal{D}_p$ is a curve in $\mathbb{C}$ passing through $x = 0$ and asymptotic to the negative real axis. Numerical computation shows:

p	$x = -1$	$x = -1 + i$	$x = -1 + 2i$	$x = -2 + 2i$
2	1.000	0.673	0.588	0.705
3	1.323	0.873	0.745	0.877
4	1.474	0.967	0.815	0.951
5	1.560	1.020	0.854	0.991

Table 5: Values of $|\lambda_{p,x}|$ near the negative real axis. The divergence region grows with p.

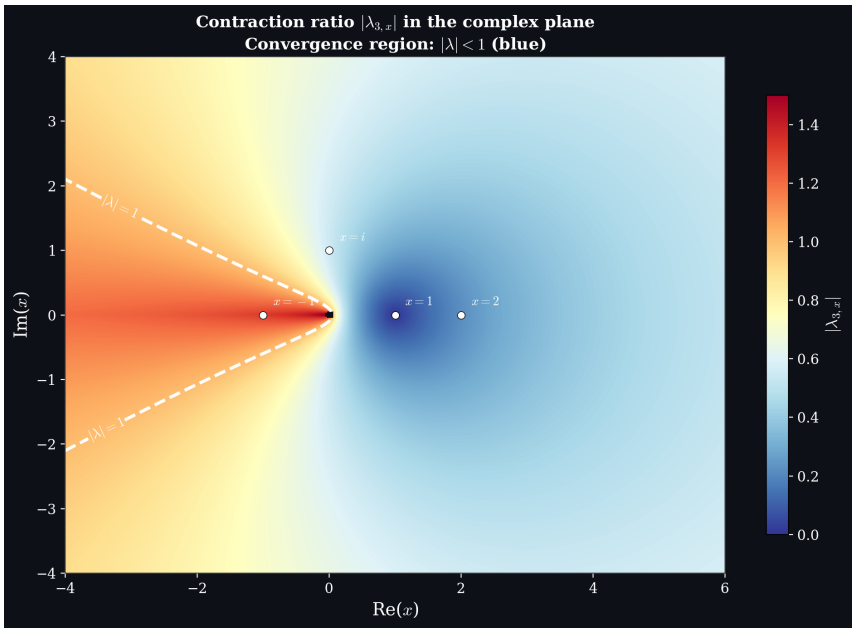


Figure 4: Heatmap of $|\lambda_{3,x}|$ in the complex plane. Blue: $|\lambda| \ll 1$ (fast convergence). Yellow/red: $|\lambda| \geq 1$ (divergence). The divergence region (red) clings to the negative real axis.

10 Conclusion

We have extended Pandrosion's geometric iteration from $\mathbb{R}^+$ to $\mathbb{C}$, establishing that:

- The complex iteration admits p **fixed points**, corresponding to the p complex p-th roots of x, equally spaced on a circle of radius $|x|^{-1/p}$.
- The contraction ratio $\lambda_{p,x}$ extends analytically to $\mathbb{C}$, with $|\lambda_{p,x}| < 1$ for all x outside a neighborhood of the negative real axis.
- The **basins of attraction** exhibit fractal boundaries, providing a new family of rational iteration fractals complementary to Newton fractals.
- **Steffensen acceleration** preserves quadratic convergence in $\mathbb{C}$, reaching machine precision in 3–5 steps for generic complex targets.
- The Riemann zeta function admits the formal representation $\zeta(s) = \prod_p S_\infty(p^{-s})$, identifying each Euler factor as a Pandrosion sum. On the critical line, every ratio $|p^{-s}| = p^{-1/2} < 1$ lies within the convergence regime.

Perspectives. Several directions merit further investigation:

- *Fractal dimension*: compute the Hausdorff dimension of the Pandrosion Julia set as a function of p and x.
- *Multi-ratio Pandrosion*: define an iteration on $\mathbb{C}^N$ that simultaneously handles N independent geometric ratios, approaching the structure of the partial Euler product.
- *Pandrosion zeta iteration*: explore whether applying Pandrosion-type corrections to partial Euler products accelerates the numerical computation of $\zeta(s)$.
- *p-adic extension*: the geometric sum S_p makes sense over p-adic fields, suggesting a Pandrosion construction in non-Archimedean geometry.

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